

A microscopic image of skeletal muscle tissue, showing multiple parallel muscle fibers. The fibers are pinkish-red and contain numerous dark blue nuclei, which are located along the periphery of the fibers. The overall structure is highly organized and striated.

Tissues II

BIOL2251

Chapter 4

Sections 4.7 – 4.12



Connective Tissue Family Tree

Connective Tissue Family Tree

- **Instructions**

- Work in small groups
- **Create a 'family tree' with branches for each type of connective tissue**
 - CT proper, supporting CT, and fluid CT
- **Under each branch, list the specific tissue types**
 - **Note 1-2 defining characteristics for each branch**
- One group presents their tree, others can add missing elements
- **Bloom's taxonomy = Understand, Analyze**

Connective Tissue Family Tree

CT Proper	Supporting CT	Fluid CT
Binds, supports, fills spaces	Structural support, protection	Transport
Loose: areolar, reticular, adipose Dense: dense regular, dense irregular Key features: Variable matrix (ground substance + fibers), fibroblasts main cell type, wide range of functions	Cartilage: hyaline, fibrocartilage, elastic Bone: compact, spongy Key features: Firm to hard matrix, supports body structures, more specialized cells (chondrocytes, osteocytes)	Blood Lymph Key features: Liquid matrix (plasma), cells suspended in fluid, transport nutrients/gases/wastes/immune cells



Cartilage Comparison Challenge

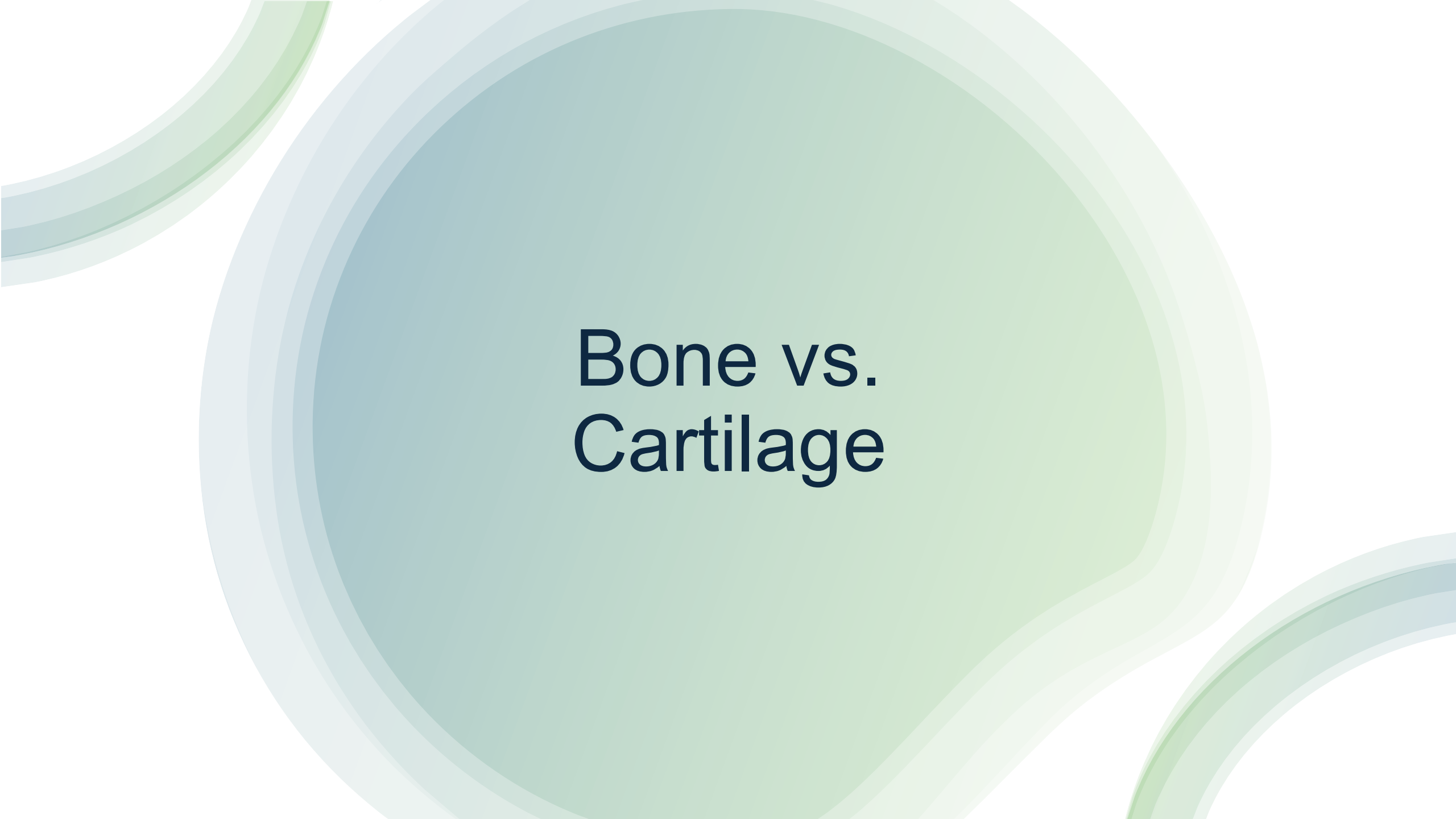
Cartilage Comparison Challenge

- **Instructions**

- Think: **Create a table comparing hyaline, fibrocartilage, and elastic cartilage**
 - Include: **matrix appearance, main fiber type, flexibility vs. strength, and one example location**
- Pair: Compare tables with partner
 - Discuss: **Why does each location need that specific type of cartilage?**
 - Share: Explain one cartilage-location match
- **Bloom's taxonomy = Understand, Analyze**

Cartilage Comparison Challenge

Feature	Hyaline Cartilage	Fibrocartilage	Elastic Cartilage
Matrix appearance	Glassy, translucent, smooth	Dense collagen fibers visible	Dense elastic fibers visible
Main fiber type	Fine collagen fibers (not visible)	Thick collagen bundles	Elastic fibers
Flexibility	Moderately flexible, some resilience	Least flexible, most resistant to compression	Most flexible, highly elastic
Strength	Moderate support	Maximum tensile strength	Less strength, high elasticity
Locations	Articular surfaces of joints, nose, trachea, ribs, fetal skeleton	Intervertebral discs, menisci of knee, pubic symphysis	External ear, epiglottis, auditory tubes
Function at location	Smooth surface for friction-free movement, supports airway	Absorbs shock/compression between bones	Maintains shape while allowing flexibility



Bone vs. Cartilage

Bone vs. Cartilage

- **Instructions**

- Groups: On whiteboard, create two columns (Bone vs. Cartilage)

Compare:

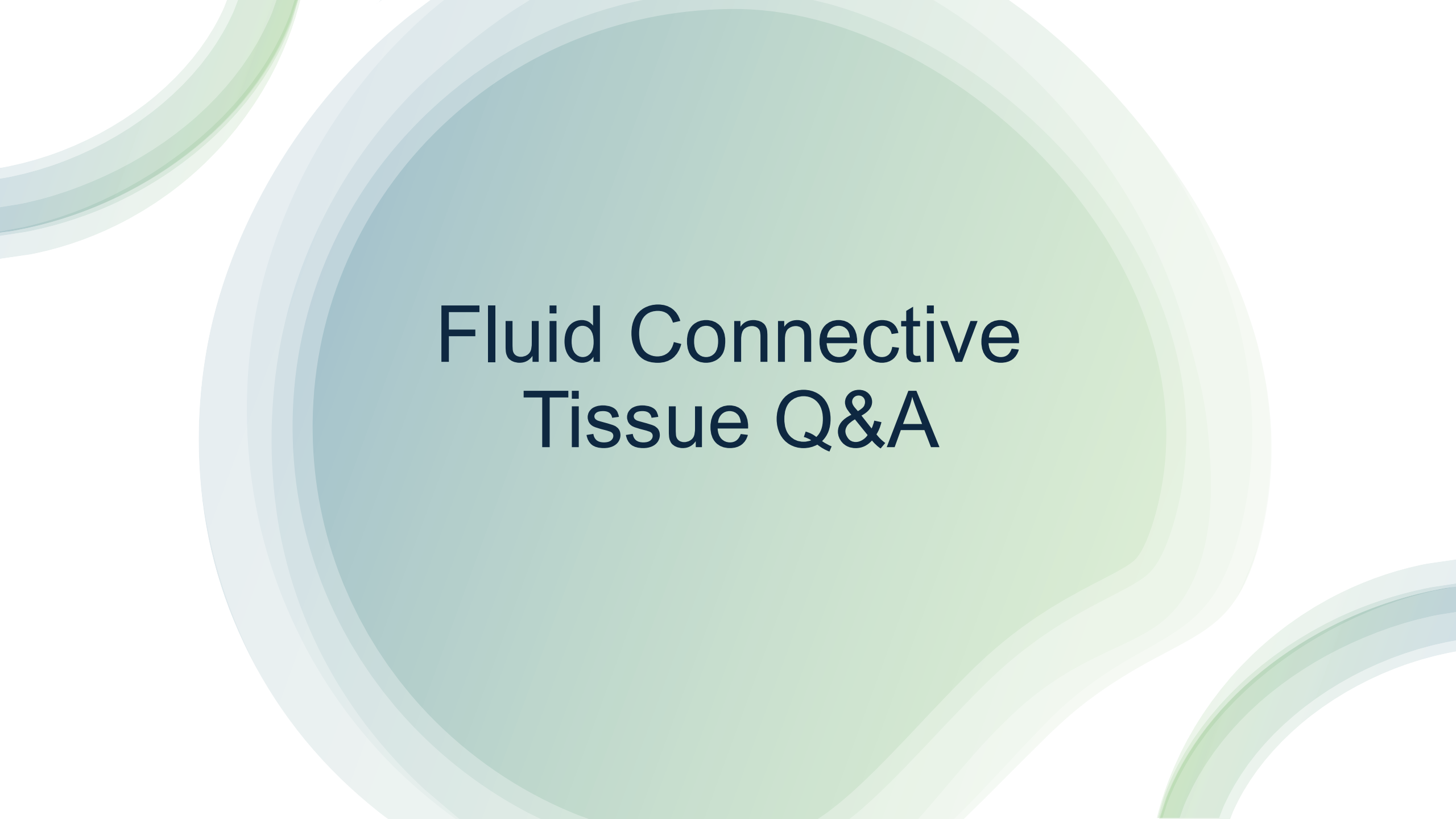
- **Matrix composition (organic vs. inorganic)**
 - **Vascularization**
 - **Cell types and their locations**
 - **Strength and flexibility**
 - **Regeneration/healing capacity**
 - **Main functions**

- Rotation: Each group will add one thing for cartilage and bone on my table on the large whiteboard

- **Bloom's taxonomy = Analyze, Evaluate**

Bone vs. Cartilage

Feature	Cartilage	Bone
Matrix	Ground substance + collagen or elastic fibers; no minerals	Ground substance + collagen fibers + calcium phosphate (hydroxyapatite crystals) = calcified/mineralized
Vascularization	Avascular (no blood vessels)	Highly vascularized (many blood vessels)
Cells	Chondrocytes in lacunae	Osteocytes in lacunae
Cell arrangement	Cells scattered in matrix	Cells in concentric rings (osteons in compact bone) or irregular arrangement (spongy bone)
Strength	Flexible, resilient, compressible	Hard, rigid, strong but can break
Flexibility	More flexible (especially elastic)	Not flexible
Regeneration	Heals very slowly (avascular)	Heals relatively well (vascularized)
Main functions	Smooth surfaces for joint movement, flexible support, shock absorption	Structural support, protection, leverage for movement, mineral storage, blood cell formation



Fluid Connective Tissue Q&A

Fluid Connective Tissue Q&A

- **Instructions**

- **I will present questions about fluid connective tissues**
 - One question presented at a time
 - **Think on your own for 30 seconds and prepare an answer**
- After 30 seconds, **discuss with a neighbor**
 - Discuss any differences and the thought process
- I will provide the correct answer, and you can self-check
- **Bloom's taxonomy = Remember, Understand**

Fluid Connective Tissue Q&A

1. What is the liquid matrix of blood called?
 2. Name the three types of formed elements in blood.
 3. What is the primary function of red blood cells?
 4. Which blood cells fight infection?
 5. What do platelets do?
1. **Plasma** (liquid portion, ~55% of blood volume, contains water, proteins, nutrients, wastes, gases, hormones)
 2. **Red blood cells (erythrocytes), white blood cells (leukocytes), platelets (thrombocytes)** (formed elements = ~45% of blood)
 3. **Transport oxygen (and some CO₂)** via hemoglobin protein
 4. **White blood cells (leukocytes)** - various types with different immune functions
 5. **Blood clotting/hemostasis** - form platelet plug, release clotting factors

Fluid Connective Tissue Q&A

6. Where is blood found in the body?

6. Within blood vessels (arteries, veins, capillaries) and heart chambers

7. What is lymph made from?

7. Excess interstitial fluid that leaks from capillaries; similar composition to plasma but less protein

8. Where does lymph flow?

8. In lymphatic vessels (lymphatic capillaries → vessels → ducts)

9. What is the main function of lymph?

9. Return excess fluid to blood, transport lipids from digestive system, immune surveillance (lymphocytes patrol for pathogens)

10. What tissue type are blood and lymph?

10. Fluid connective tissue - liquid matrix with cells suspended in it

The background features decorative curved lines in shades of green and blue, sweeping across the top and bottom of the slide.

Muscle Tissue Matching

Muscle Tissue Matching

- **Instructions**

- Work in small groups
- I will provide a **list of 18 statements about muscle characteristics, functions, and locations**
- You will create a table with three columns
 - **Skeletal, Cardiac, Smooth**
- In your groups, **sort statements into the correct column**
 - Some statements could be sorted into multiple columns
- First group to submit gets their table reviewed
 - **All correct answers = group wins a prize**
- **Bloom's taxonomy = Remember, Understand, Apply**

Muscle Tissue Matching

Statements to Sort		
A. Striated appearance under microscope B. Voluntary control C. Involuntary control D. Single central nucleus per cell E. Multiple nuclei per cell (periphery) F. Branching cells with intercalated discs	G. Long, cylindrical, unbranched cells H. Spindle-shaped cells I. Found in walls of hollow organs J. Attached to bones K. Found only in the heart L. Controls movements of skeleton	M. Pumps blood N. Moves substances through organs (peristalsis) O. Gap junctions allow coordinated contraction P. Moves rapidly and powerfully Q. Contracts rhythmically without nervous stimulation R. Slowest contraction, can sustain tension

Muscle Tissue Matching

Skeletal	Cardiac	Smooth
• A (striated - organized sarcomeres) • B (voluntary - conscious control) • E (multinucleated, nuclei at periphery) • G (long cylindrical fibers, unbranched) • J (attached to bones via tendons) • L (moves skeleton - locomotion, posture) • P (rapid, powerful contractions; fatigues quickly)	• A (striated - organized sarcomeres) • C (involuntary - automatic) • D (typically one central nucleus, sometimes two) • F (branching cells connected by intercalated discs with desmosomes and gap junctions) • K (found ONLY in heart wall) • M (pumps blood throughout body) • O (gap junctions allow cells to contract as unit) • Q (autorhythmic - intrinsic pacemaker, doesn't need nerve signal)	• C (involuntary - automatic/autonomic control) • D (single central nucleus) • H (spindle-shaped, tapered ends) • I (walls of hollow organs - stomach, intestines, bladder, uterus, blood vessels, airways) • N (peristalsis - waves of contraction move substances) • O (gap junctions coordinate contraction) • R (slow contraction, resistant to fatigue, can maintain tension)



Neuron Structure and Function Mapping

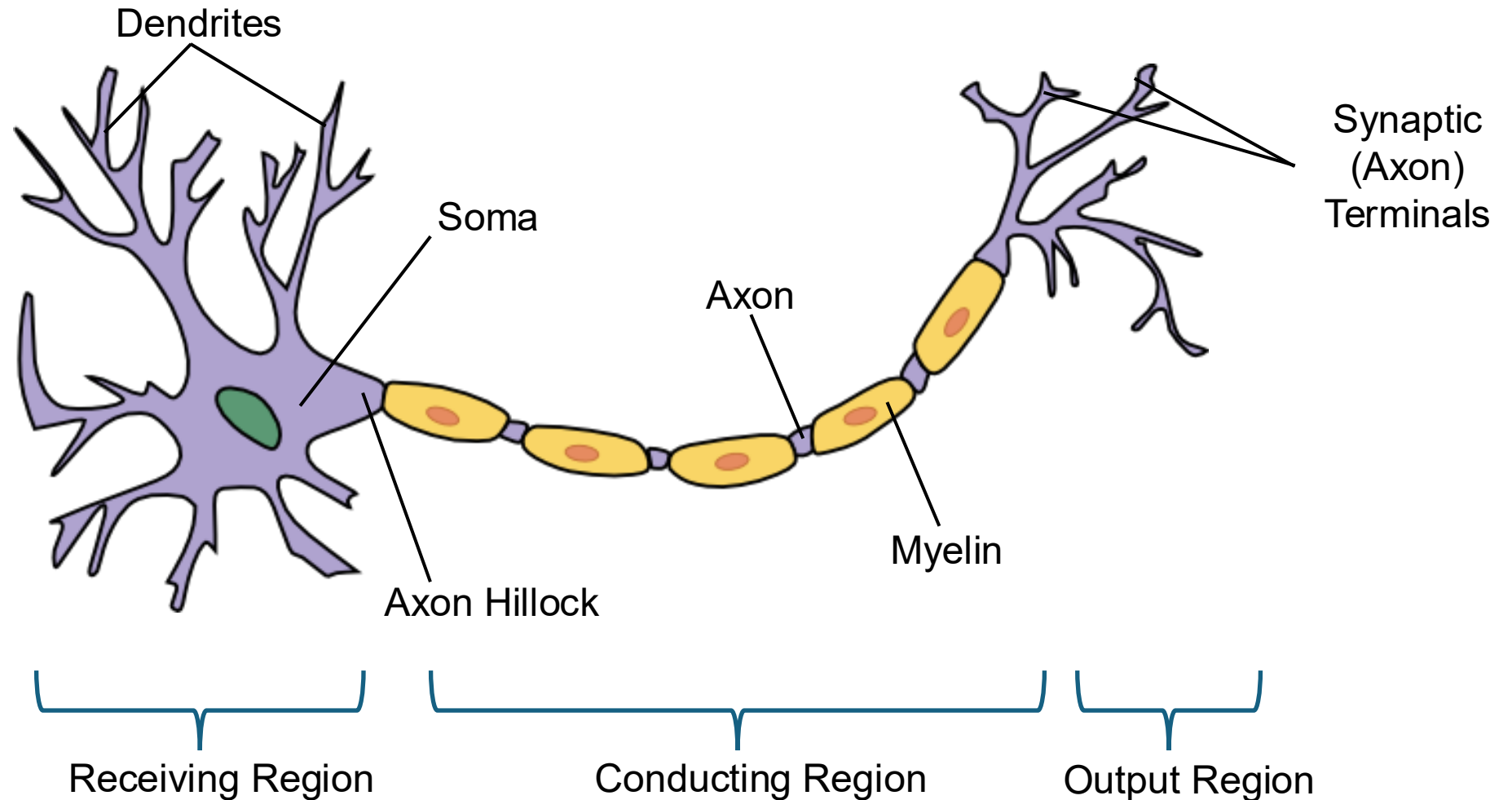
Neuron Structure and Function Mapping

- **Instructions**

- Think: **Draw a neuron and label: dendrites, cell body (soma), axon, axon/synaptic terminals**
 - **Classify each structure as receiving region, conducting region, or output region**
- Pair: Compare drawings with partner
- Share: Class discussion - **How does information flow through a neuron?**

- **Bloom's taxonomy = Understand, Analyze**

Neuron Structure and Function Mapping





Membrane Madness Mini- Cases

Membrane Madness Mini-Cases

- **Instructions**

- Work in small groups (12 total)
- Each group will receive 2 scenarios
- For each scenario:
 - **Identify the membrane type**
 - **Mucous, serous, cutaneous, or synovial**
 - **Explain what tissues compose it**
 - **Describe its function in this location**
- Presentations: Each group shares one scenario, class votes on agreement
- **Bloom's taxonomy = Apply, Analyze**

Membrane Madness Mini-Cases

Scenario 1: A patient with a respiratory infection is coughing up thick mucus. The doctor explains that the lining of the respiratory tract is producing extra secretions to trap and expel pathogens.

Answer: Mucous membrane

- **Tissues:** Epithelium (varies by location - pseudostratified columnar in respiratory) + underlying areolar CT (lamina propria)
- **Function:** Lines body cavities open to exterior; secretes mucus for protection, lubrication, and trapping pathogens
- **Other locations:** Digestive, urinary, and reproductive tracts

Scenario 2: During abdominal surgery, the surgeon notices a smooth, moist membrane covering the intestines and lining the abdominal cavity. This membrane produces a slippery fluid that allows organs to move against each other without friction.

Answer: Serous membrane

- **Tissues:** Simple squamous epithelium (mesothelium) + thin areolar CT
- **Function:** Lines closed body cavities; secretes serous fluid (watery, lubricating) to reduce friction between organs
- **Layers:** Parietal (lines cavity wall) and visceral (covers organs)
- **Other locations:** Pleura (around lungs), pericardium (around heart), peritoneum (abdominal cavity)

Membrane Madness Mini-Cases

Scenario 3: A dermatologist examines a patient's skin, noting that this protective barrier prevents water loss and blocks pathogen entry while also providing sensory information about the environment.

Answer: Cutaneous membrane

- **Tissues:** Stratified squamous epithelium (epidermis - keratinized) + dense irregular CT (dermis)
- **Function:** Covers body surface; protects against dehydration, pathogens, UV radiation, mechanical damage; temperature regulation; sensory reception
- **Unique feature:** Only DRY membrane (others are moist); most exposed membrane

Scenario 4: An orthopedic surgeon injects lubricating fluid into a patient's painful knee joint. The membrane lining the joint cavity produces this fluid to reduce friction between the bones during movement.

Answer: Synovial membrane

- **Tissues:** Specialized connective tissue lining (NO epithelium - unique!) + areolar/adipose CT
- **Function:** Lines joint cavities; secretes synovial fluid (viscous, lubricating) to reduce friction, nourish articular cartilage, and cushion joints
- **Other locations:** All freely movable (synovial) joints - shoulder, hip, elbow, knee, etc.
- **Unique feature:** ONLY membrane with no epithelial layer

Membrane Madness Mini-Cases

Scenario 5: A patient with pleurisy experiences painful breathing because inflammation has reduced the lubricating fluid between the lung and chest wall, causing these surfaces to rub together.

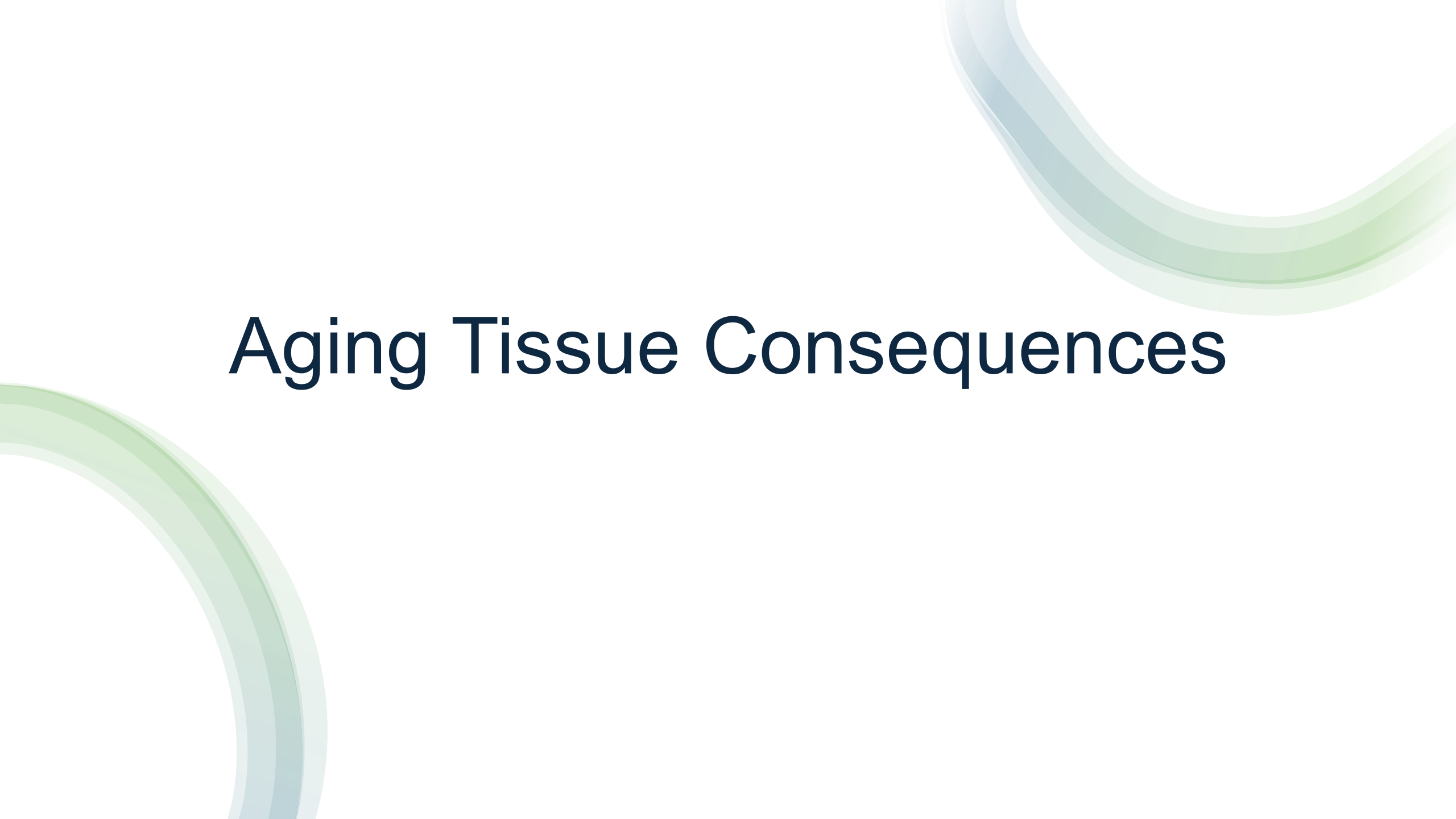
Answer: Serous membrane

- **Tissues:** Simple squamous epithelium + areolar CT
- **Function:** Visceral pleura covers lungs, parietal pleura lines chest wall; pleural fluid between layers allows smooth expansion/contraction during breathing
- **Problem in pleurisy:** Inflammation reduces fluid, increases friction

Scenario 6: A patient with acid reflux has damage to the lining of the esophagus. The doctor explains that stomach acid is eroding the protective mucus layer that normally shields the esophageal tissue.

Answer: Mucous membrane

- **Tissues:** Stratified squamous epithelium (in esophagus - needs protection from abrasion) + lamina propria (areolar CT)
- **Function:** Mucus protects epithelium from mechanical damage and chemical damage (though stomach acid can still harm if excessive)
- **Why it matters:** Chronic exposure can lead to ulcers or Barrett's esophagus

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Aging Tissue Consequences

Aging Tissue Consequences

- **Instructions**

- Work in small groups (4-6 people)
- **Each group will be assigned one tissue type**
- Discuss and prepare a brief presentation
 - **How does this tissue change with aging?**
 - **What health problems result from these changes?**
 - **Why does the structural change cause a functional problem?**
 - **Can anything slow or prevent these changes?**
- Presentations: Each group shares for their tissue type
- **Bloom's taxonomy = Analyze, Evaluate**

Aging Tissue Consequences

- **Decreased regenerative capacity** - slower cell division and repair
- **Accumulation of damage** - oxidative stress, cross-linking of proteins
- **Reduced metabolic activity** - less ATP production, slower processes
- **Increased fibrous tissue** - scarring replaces functional tissue
- **Loss of cells** - especially in non-dividing tissues (muscle, nerve)



Tissue Integration

Tissue Integration

- **Instructions**

- Work on your own
 - **You will solve a physiological scenario that requires knowledge of multiple tissue types and how the tissues work together**
 - **Submit your work before leaving class**
 - Make sure you have completed the full worksheet and added your name
- **Bloom's taxonomy = Apply, Analyze, Evaluate**